

DisruptionPy: An open-source physics-based scientific framework for disruption analysis of fusion plasmas

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Summary

DisruptionPy (Rea et al., 2024; Trevisan et al., 2024, 2026; Wei et al., 2024) is an open-source physics-based scientific framework for disruption analysis of fusion plasmas, designed with the explicit purpose of streamlining database preparation of experimental fusion data to allow efficient Artificial Intelligence and Machine Learning (AI/ML) workflows.

DisruptionPy originated as an institutional effort from the Plasma Science and Fusion Center within the Massachusetts Institute of Technology (MIT PSFC) to create a shared and validated set of feature-extraction routines, and evolved into an open-source scientific framework in order to aid disruption scientists everywhere. DisruptionPy natively supports efficiently extracting data from MDSplus (Stillerman et al., 1997, 2025), the leading open-source storage back-end for most fusion experiments, and enables scientists to carry out complicated Python-based computations at scale across entire experimental databases. DisruptionPy also supports extracting data using Xarray, which is interoperable with the open MAST (Jackson et al., 2024, 2025) dataset, therefore enabling researchers to easily access and analyze historical MAST data without the need to participate in a collaboration agreement. DisruptionPy relies on established numerical libraries, e.g. NumPy, SciPy, Pandas, Xarray, to allow effortless manipulation of either raw or pre-processed data into complicated feature-extraction workflows for database generation.

The heterogeneous set of scripts from which DisruptionPy was developed led to several high-profile scientific publications (Hu et al., 2021; Montes et al., 2019; Rea et al., 2018, 2019, 2020; Rea & Granetz, 2018; Tinguely et al., 2019; J. Zhu et al., 2021; J. X. Zhu et al., 2020, 2023). DisruptionPy itself, as the new high-quality basis for the scientific work of the entire Disruptions Group at MIT PSFC, enabled further publications in recent years (Keith et al., 2024; Maris et al., 2024; Saperstein et al., 2025; Spangher et al., 2025) and will undoubtedly continue to lead to further high-impact results in the near future, as more scientific projects that leverage it reach maturity.

Statement of need

Magnetically-confined fusion experiments routinely operate under a wide variety of engineering parameters in order to gain invaluable insight into fusion plasmas with the purpose of understanding and then harnessing their intrinsic power for energy production. Such exploration of a wide parameter space sometimes results in unexpected and rapid loss of confinement of the plasma discharge, events which are generically known as ‘disruptions’. Disruptions represent a significant danger to both modern experimental machines and, above all, future

reactor-relevant devices. Therefore preventing disruptions, detecting them, and avoiding them are features of paramount importance for any plasma control system. Given the sheer number of available diagnostic systems, and possible plasma modeling tools, artificial intelligence and machine learning (AI/ML) models are ideal candidates for heavy-duty numerical computation. Fast and agile numerical frameworks for database preparation and preprocessing are necessary for letting researchers focus on novel algorithms and benchmark different architectures and models.

As the fusion community prepares for the upcoming burning plasma devices, the multiple existing data repositories already face numerous interoperability challenges. Previous community reporting (Humphreys et al., 2020) identified the need to improve several aspects of existing platforms, ranging from hardware and technology to software, including development of optimized ML-ready workflows for scientific discovery. Therein, the authors highlighted the current different data access systems, the various data storage formats, and a lack of adequately-labeled data as main challenges that need to be addressed by the research community.

In such context, the open-source development of DisruptionPy satisfies the crucial need for shared and validated data-processing workflows, and the framework's helpfulness will only grow as more experimental devices relax their requirements for data access and evolve towards open data and FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson et al., 2016).

Additional examples of similar frameworks for experimental data retrieval and database preparation are TokSearch (Sammuli et al., 2018) and DEFUSE (Pau et al., 2023). The TokSearch library (Sammuli et al., 2018) was developed to efficiently query, process, and analyze experimental data from DIII-D for ML applications. It leverages a distributed file format to increase throughput and a dedicated API to transfer data from MDSplus and export it in Parquet format. TokSearch appears to be established only for DIII-D workflows. DEFUSE (Pau et al., 2023), the Disruption and Event analysis framework for FUSion Experiments, implements an interface layer to access the data from different fusion experiments through MDSplus and HDF5. Source data, diagnostics, machine descriptions, and data-processing schemes are defined in interoperable data libraries in JSON format within a data abstraction layer. DEFUSE has been applied to several devices, however the framework has not been open-sourced yet.

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